ENTERPRISE ARCHITECTURE SPECIFICATION

System Requirements Specification (SRS)

Enterprise Local-First Desktop RPA & Autonomous Agent Engine

Cross-Platform Orchestration with Human-In-The-Loop Quality Control

Document ID:	SRS-RPA-2026-V1.0
Target Baseline:	Tauri v2 + React 18 + Python 3.11 LangGraph + Gemini API
Classification:	Enterprise Local-First (Zero Cloud Database, Strict BYOK)
Author / Role:	Principal Systems Architect & Lead Full-Stack Engineer
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Status:	Approved & Verified Baseline Release

■ ARCHITECTURAL CERTIFICATION STATEMENT

This software engine satisfies all five baseline invariants: (1) Local-First compute with no server database, (2) Direct BYOK Gemini communication, (3) LangGraph interrupt-driven HITL pause before external state mutations, (4) Zero-CLI automated dependency provisioning, and (5) Non-hijacking detached Playwright browser execution.

Document Revision History

Version	Release Date	Author / Role	Primary Changes & Scope
0.1.0	2026-10-24	Systems Architect	Initial SRS draft and core architectural constraints definition.
0.9.0	2026-10-24	Lead Full-Stack Eng	Integration of Tauri IPC bridge, LangGraph state machine & Playwright CDP.
1.0.0	2026-10-24	Principal Architect	Final verification: zero-CLI auto-provisioning, native OS keyring & vision fallback.

Executive Summary

This System Requirements Specification (SRS) defines the formal requirements, topological structure, operational workflows, and compliance bounds for the Enterprise Local-First Desktop RPA Platform. Engineered to replace bloated, cloud-dependent automation tools, this system executes browser automation, data extraction, and multi-step desktop RPA on the user's workstation without relying on third-party backend servers.

1. Introduction & Architectural Guarantees

The system enforces five architectural invariants that govern all data flow, process execution, and network activity:

■ GUARANTEE 1: STRICT LOCAL-FIRST COMPUTE

Zero cloud databases, telemetry proxies, or server-side execution queues. All execution (browser control, filesystem operations, SQLite WAL state persistence) occurs locally on the client host.

■ GUARANTEE 2: BRING YOUR OWN KEY (BYOK)

Direct client-to-API communication with Google Gemini API (generativelanguage.googleapis.com) over TLS 1.3. API keys are stored in the OS Keyring (macOS Keychain / Windows Credential Manager) or local AES-256-GCM vault.

■ GUARANTEE 3: DRAFT-FIRST QUALITY CONTROL (HITL)

Agents MUST NOT perform destructive external state updates without hitting a Human-in-the-Loop pause step powered by LangGraph state checkpointers and interrupt() logic.

■ GUARANTEE 4: ZERO-CLI AUTO-BOOTSTRAP

Dynamically inspects runtime availability (Python 3.11+, Playwright drivers, pip) and auto-provisions required binaries to sandboxed user directories without root or admin prompts.

■ GUARANTEE 5: HEADLESS & DETACHED AUTOMATION

Web automation runs in detached Playwright/CDP instances. The engine never hijacks active system mouse cursors or steals window focus.

2. System Architecture & Component Model

The platform is divided into three distinct runtime layers communicating across decoupled local IPC channels:

System Layer	Technology	Core Functional Scope	Interface
Desktop Shell	Tauri v2 (Rust)	Host windowing, OS Keyring bridge, SQLite WAL manager, pre-flight checks	Native FFI / OS APIs
IPC Bridge	Tauri Core IPC	Asynchronous command dispatch and bidirectional JSON-line stdio pipes	JSON over Stdio
Frontend UI	React 18 + Tailwind	Chat intake, execution log viewing, HITL preview modal, Settings	Zustand Reactive Store
Execution Sidecar	Python 3.11 + LangGraph	StateGraph engine, ethical filter, Playwright CDP, Vision Fallback	JSON-line Stdio
AI Provider	Google Gemini 2.0 Flash	Intent classification, manifest planning, multimodal vision locator	HTTPS / Direct BYOK

3. Functional Requirements Specification

3.1 Security, Storage & Pre-Flight Intake (Phase 1)

ID	Requirement	Technical Specification
FR-SEC-01	Native OS Keyring	Store Gemini API keys in the native OS Keyring under service 'rpa-agent-platform'.
FR-SEC-02	AES-256 Vault Fallback	When keyring is unavailable, encrypt keys with AES-256-GCM using machine SHA-256 derivation.
FR-SEC-03	Regex Blocklist Filter	Intercept prohibited attack patterns (credential theft, DDoS, ransomware) prior to LLM evaluation.
FR-SEC-04	Gemini Ethical Classifier	Evaluate prompt safety with Gemini 2.0 Flash; reject unsafe tasks with structured explanation.
FR-SEC-05	Structured Task Manifest	Decompose valid prompts into atomic TaskManifest JSON specifying tools, schemas, and dependencies.

3.2 Runtime Engine & Resiliency (Phase 2)

ID	Requirement	Technical Specification
FR-ENG-01	LangGraph StateGraph	Orchestrate agent state flow through modular nodes: intake, plan, execute, review, revise, finalize.
FR-ENG-02	SQLite WAL Checkpointing	Persist graph execution checkpoints to SQLite to enable zero-loss state recovery across crashes.
FR-ENG-03	Playwright CDP Browser	Automate web tasks via detached Playwright headless instances without cursor hijacking.
FR-ENG-04	Multimodal Vision Fallback	Upon DOM selector failure, capture viewport screenshot and locate element via Gemini Vision.
FR-ENG-05	Circuit Breaker Protection	Limit self-healing retry loops to max 3 attempts or 5 minutes (300s) before auto-tripping.
FR-ENG-06	Sandboxed Filesystem	Restrict filesystem operations to local sandbox workspace with path traversal enforcement.

3.3 Interactive Review & HITL Quality Control (Phase 3)

ID	Requirement	Technical Specification
FR-HITL-01	LangGraph Interrupt Pause	Halt execution before draft finalization or external state modifications using interrupt().
FR-HITL-02	Interactive Preview Modal	Render generated drafts (Text, PDF, CSV, JSON) with syntax highlighting and line numbers.
FR-HITL-03	Approval Sign-Off	Support one-click approval dispatching decision='approve' to trigger finalize node.
FR-HITL-04	Incremental Diff Updates	Submit user change requests to Gemini with target sections only to conserve token window.

4. External Interfaces & Stdio Protocol

Communication between the Tauri desktop frontend and Python sidecar uses newline-delimited JSON messages:

Stream	Message Type	Payload Data Structure	Operational Trigger
Tauri → Python	start	{ prompt: str, api_key: str }	Initiates new agent execution thread
Tauri → Python	resume	{ thread_id: str, decision: 'approve'|'revise', feedback: str }	Resumes paused LangGraph checkpoint
Tauri → Python	cancel	{ thread_id: str }	Cancels active execution thread
Python → Tauri	status	{ step: str, message: str }	Emits StateGraph transition updates
Python → Tauri	safety_result	{ classification: 'safe'|'unsafe', reason: str }	Emits pre-flight ethical classification
Python → Tauri	manifest	{ task_id: str, steps: [...], ... }	Emits generated execution plan manifest
Python → Tauri	hitl_pause	{ draft_id: str, file_type: str, content: str }	Signals interrupt() pause for operator review
Python → Tauri	draft_updated	Updated DraftOutput object	Pushes incremental revision draft to UI
Python → Tauri	completed	{ draft: object, message: str }	Notifies task completion

5. Non-Functional Requirements

ID	Category	Requirement Statement
NFR-SEC-01	Confidentiality	Zero telemetry proxies; no payload leaves the machine except direct TLS 1.3 calls to Gemini.
NFR-SEC-02	Cryptographic Rigor	Local vault fallback enforces authenticated AES-256-GCM with unique random nonces.
NFR-PERF-01	Responsiveness	UI interactions and streaming status badges update at 60 FPS (latency < 16ms).
NFR-PERF-02	Cold Startup	Application cold startup to interactive chat intake completes in less than 2.0 seconds.
NFR-REL-01	Crash Recovery	System crashes allow 100% thread resumption from persistent SQLite WAL checkpoints.
NFR-REL-02	Circuit Timeout	Self-healing loops trip after max 3 retries or 300 seconds timeout to prevent hanging.

6. Verification & Traceability Matrix

Requirement	Verification Method	Test Procedure	Compliance
FR-SEC-01..02	Unit & System Test	Validate key storage in macOS Keychain & fallback to AES-256-GCM vault.	VERIFIED PASSED
FR-SEC-03..04	Integration Test	Execute blocked attack vectors; assert sub-millisecond rejection.	VERIFIED PASSED
FR-ENG-01..02	Graph Pipeline Test	Stream LangGraph StateGraph nodes; verify SQLite WAL checkpointing.	VERIFIED PASSED
FR-ENG-04	Vision Locator Test	Simulate selector timeout; verify viewport screenshot & coordinate extraction.	VERIFIED PASSED
FR-HITL-01..04	End-to-End Test	Assert interrupt() pause, modal preview, revision loop, and approval sign-off.	VERIFIED PASSED
FR-BOOT-01..03	Bootstrap Test	Verify zero-CLI auto-provisioning of Node.js, Python 3.11, and Playwright.	VERIFIED PASSED